
SCALING ONCOLOGY LLM GENERATED CODE: JAN 2026

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*Two page guide for oncology LLM code generation and rapid prototyping at scale
Optimizations and bug-fixes may be required throughout multi-prompt processes*

Prompting: 500-1500 Line Outputs

- For scaling with proprietary models: avoid submitting the same prompt twice to mitigate failed code generations (even if prompts are in different conversations). LLMs reward novel and well defined instruction over repetitive and small requests [1, 2].
- Improve code generation quality in performant LLMs by only including a single upload in a prompt (notebook, script, pdf, website). Combine information into one file where possible to maximize output quality and lines of code.
- Optimize context length utilization by decreasing unnecessary content from being introduced with added prompts in a conversation. Important: subsequent prompts must be as detailed as the first prompt to reduce chances of code truncation.
- Research and Deep Research types of additional features do not reliably generate code at scale. These options should primarily only be used to generate data or build knowledge from online sources regarding a topic of interest [3, 4].
- To create code used to process data files, provide LLM with code instructions and 5-10 row samples of each data file. Use the LLM and data samples to create the processing script. The generated code with the full dataset should be processed by IDEs.
- Charts/diagrams a) Prompt LLM with dataset to receive Python visualization scripts (matplotlib, plotly). Quality screening and re-positioning may be required for some scripts to obtain high quality images. Direct chart image requests are limited [5, 6].
- Charts/diagrams b) Meta prompt LLM with uploaded dataset to obtain visualization instruction prompt. 2nd prompt: Paste instruction prompt with same dataset to obtain ≤ 10 Python visualization scripts using effective LLMs [7, 8].

Oncology Trial LLM Implementation

- Utilize LLMs where Phase I/II/III time and cost efficiency improvement potential is highest. Oct 2025 paper [9]: Inputs 25+ author LLM applications, current trial activities, and regulatory standards; Outputs LLM Δ Time and LLM Δ Cost potential
- LLM Quantitative Systems Pharmacology
 - Sanofi 2024: QSP modeling with AI for virtual Phase 1b trials Δ Time -2 to -3 months, Δ Cost -\$200K to -\$800K [10]
 - Certara 2025: AI-enabled QSP platform & purpose-built GPT Δ Time -1 to -1.5 months, Δ Cost -\$20K to -\$40K [10, 11]
 - Kawchak 2025: Pancreatic cancer QSP, VVUQ in weeks Δ Time -2 to -3 months, Δ Cost -\$50K to -\$110K [11, 12]
- LLM Generated Digital Twin Simulations
 - Sanofi 2024: Dose optimization through digital twin simulations Δ Time -1.5 to -2 months, Δ Cost -\$40K to -\$80K [10]
 - Vidovszky 2024: AI-generated digital twins using PROCOVA Δ Time -0.8 to -1.2 months, Δ Cost -\$16K to -\$35K [13, 14]
 - Kawchak 2025: FDA compliance hastened: PDAC digital twin Δ Time -10 to -15 months, Δ Cost -\$2.5M to -\$4M [15, 16]
- Oct 2025 Guidance Yields New Innovation
 - Kawchak Nov 25: LLM-Generated glioblastoma clinical trial patient matching ML, 6 full FDA/ICH/ISO documents [17]
 - 1) AI Peer review: Multi-LLM iterations and corrections conducted in middle of manuscript development process
 - 2) AI Recommendation: ML model efficiency change resulted in test set accuracy improvements from 67% to 94%
 - 3) Human, iteration, timeline & cost efficiencies standardized. 0.831 Rating, with minor optimization opportunity [18]
 - Kawchak Dec 25: 16 Proprietary vs. open-source LLMs code generation goal-based competition on FDA FAERS [19]
 - 1) Performant LLM created, optimized, and bug-fixed head-to-head code competition notebook with human-in-the-loop
 - 2) Iterative learning of up to +0.405 score per round: Correctness, methodology, quality & algorithm efficiency metrics
 - 3) Tournament champion improved from first round, and maintained top score for three additional rounds of 0.9775
- Jan 2026 LLM Code Generation Goals
 - Implementation cycle: Performant new code generation LLMs incorporated into current workflow within 30 days [20]
 - Evaluate new proprietary and open-source LLMs head-to-head for workflow appropriateness every implementation cycle
 - Max Efficiency: $n = 1$. Efficiency = $1/n$. n = number of humans, n = number of prompts, n = total cost, n = time taken [17]

Cite This Work

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